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RESEARCH ARTICLE

# A Robust Hybrid CNN+ViT Framework for Breast Cancer Classification Using Mammogram Images

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**ABSTRACT** Breast cancer is the most frequent type of cancer largely experienced by women currently, although it could happen to men also. It appears when abnormal breast tissue cells grow rapidly and form tumors. Mammogram is a technique that is employed by doctors to analyse the breast in the diagnosis of early cancer. These mammograms are classified into Benign and Malignant. This research addresses the variability and potential oversight in radiologists' manual mammogram interpretations, aiming to enhance classification accuracy by combining Convolution Neural Networks (CNNs) and Vision Transformers (ViTs). CNN is a successful image classification that uses hierarchical feature extraction, ViTs capture the global context but require substantial data and computation. In this research, we have used CLAHE-enhanced mammogram images from Kaggle for training and applied a CNN+ViT model. We have also used a few pre-trained models such as DenseNet, Inception, SE Resnet, and XceptionNet for comparative analysis. The CNN+ViT model gave us an accuracy of 90.1% showing robust performance. Although XceptionNet achieved perfect accuracy, it may indicate overfitting.

**INDEX TERMS** Breast cancer, mammogram screening, CNN, ViT transformers, hybrid model, local features, global features.

## I. INTRODUCTION

In 2024, breast cancer will be considered the most life-threatening and prevalent one among women globally. In the tissues of the breast, this cancer develops and shows symptoms when breast cells abnormally grows and forms a lump or mass, and compared to cells that ae healthy, these cells growth increases rapidly [1]. Like this, the proportion of these cells increases in breast cancer. The Breast Cancer Care (BCC) survey states that 42% of NHS trusts do not have enough experienced specialists in breast cancer; it is the cause for the decreasing rate of survival of breast cancer patients [2], [3]. A study by the World Health Organization (WHO) says that, of all affected patients, only 8% are diagnosed, 6% have passed away, the remaining people are not diagnosed, and most of them are females [4], [5].The mortality rate of breast cancer reaches 627000 deaths, while India amounts to 14.0% of total cases beginning at early thirties [6], [7].

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Early detection and diagnosis remain key to improving the long time survival rate of breast tumor disease. Nowadays, the techniques of medical imaging, ultrasound, and mam-mography are widely used for the detection of breast cancer [8]. Compared to other mammography techniques, it is a less painful and non-invasive screen technique because it can capture x-ray images with all tissue details [9]. Mammograms help in finding the potential abnormalities, which are classified into two categories: benign and malignant [10]. Benign tumors are harmless; they are often oval or round with smooth edges, while malignant tumors are cancerous and have the potential to grow or spread rapidly; they are often irregular and have spiky or ill-defined edges [1], [11].

In spite of the greatness of mammography in screening, mammography interpretation still faces challenges due to tissue density variations, and distinguishing benign and malignant in the early stages is very difficult. To solve this problem, we need advanced technologies to create diagnostic tools to assist radiologists to give more accurate assessments and reduce rates of misdiagnoses and for improving self- examination practices in women [12]. In the last few years, expertized systems have been introduced for breast cancer diagnosis using mammogram image datasets; they are falling into two main types: (i) image processing conventional techniques [13] and (ii) diagnosis based on deep learning [14], [15].

According to our related works, various techniques are used to solve this mammography problem. For example, the BC2NetRF framework utilizes EfficientNet-b0 with HRLG contrast enhancement, while other studies integrate EfficientNet-B4 and Equilibrium-Jaya controlled algorithms for feature selection and transfer learning, respectively [2]. Techniques like YOLO and U-Net models are frequently applied for lesion segmentation [8], with CNN architectures like VGG-11, ResNet, and DenseNet further optimized through fuzzy ensemble approaches [16], [17]. Some research explores ViTs for image classification due to their global context capture, which is enhanced through pre-training on large datasets [18], [19], [20]. In spite of their achievements, some challenges are there, including generalization due to data diversity and scarcity, computational complexity, and models overfitting when we use CNN or ViTs independently [21].

The objective of our work is to create an efficient model using the CNN+ViT model to address some of the abovementioned challenges [22], [23]. We want to improve efficiency in classification and detection by using a CNN+ViT model, which includes both CNN and ViT models. So that we can leverage both models abilities, such as CNN's ability of extraction of features locally and ViT's capability in context capturing globally [24]. We want to develop more accurate diagnostic tools with this CNN+ViT model. We use CLACHE enhanced mammogram images for getting clarity features in the dataset. Unlike the previous works, our CNN+ViT model can get balanced feature extraction and yield good classification accuracy without over fitting. Finally, paving the way for an interpretable and robust model for diagnosing breast cancer by detecting the benign and malignant tumors. We organized this paper with 5 sections excluding abstracts, keywords, and references to give a brief idea about our work. Section I gives the intro of the problem with some statistical analysis and about the background study. Section II tells about a literature survey and talks about already existing technologies and their achievements and challenges. Section III is all about the flow of information for the proposed solution. Section IV and Section V give the proofs visually regarding results and conclude the manuscript.

## II. RELATED WORK

These studies show the recent advancements in the various techniques and models of deep learning for the detection and classification of breast cancer using mammogram images. Researchers developed frameworks by using EfficientNet architecture, Equilibrium-Jaya, Mask R-CNN, and U-Net for efficiently classifying benign and malignant tumors. Some of them integrated the YOLO models with Vision Transformers and Eigen-CAM saliency maps to optimize feature extraction. They work on fuzzy ensembles of CNNs like VGG-11 and Inception v3 to improve the efficiency and performance of the model.

An automated CAD system for the detection and classification of mass in both CESM and FFDM images is developed in YOLO-Based CAD Framework with ViT Transformer for Breast Mass Detection and Classification in CESM and FFDM Images by Hassan et al. [1]. This CAD framework uses YOLOv4 for mass detection, paired with a ViT transformer for classification, and has achieved notable automation in CESM mass detection and classification tasks. The model attained classification accuracy rates of 95.65% on the INbreast dataset and 97.61% on CESM dataset.

The BC2NetRF framework for classification of breast cancer from mammogram images, proposed by Jabeen et al. [2], achieved accuracies of 95.4% and 99.7% on the CBIS-DDSM dataset and the INbreast dataset. They used the EfficientNet-b0 model for feature extraction, and for improving image clarity, they used HRLG Contrast Enhancement. They integrated the Equilibrium-Jaya Controlled Regula Falsi algorithm to improve performance. They want to work with larger and more diverse clinical datasets in the future.

Chakravarthy et al. [4] present a model for classification of breast tumor with enhanced transfer learning and chaotic map-based optimization techniques for selection. They obtained the accuracies of 98.4% and 96.1% on the INbreast and the CBIS-DDSM datasets by using the EfficientNet-B4 architecture for classification of breast cancer. In the future, they want to work with multimodal data fusion with explainable AI.

Wang [8] proposed model using mammograms along with deep learning frameworks for detection of breast cancer, which mainly aimed at enhancing accuracy in mammography for breast cancer screening. Their study showcases the usage of Mask R-CNN and U-Net models for breast lesion segmentation for effectively classifying malignant and benign tumors with an accuracy of 0.88%. In their future research, they want to improve accuracy by optimizing computational efficiency.

In their study Prinzi et al. [9] explored the YOLO architecture along with Eigen-CAM saliency maps for breast cancer detection in mammograms. Their model focuses on providing an explainable system with good accuracy, which can support physicians decisions during cancer screening programs. They achieved a map of 0.621 with a small YOLOv5 model. In the future they want to work on large datasets with less complexity models.

Mohammed and Ekmekci [14] introduce a breast cancer diagnosis system using an architecture that is a combination of YOLO-based multiscale parallel CNN and Flattens threshold frameworks. To enhance the feature extraction, they include specialized CNN components like PFES, DCB, and IB. With accuracies of 98.7% and map of 91.1% they obtained better performance. They want to explore transfer learning and adaptation of domains for boosting generalization.

Chakravarthy et al. [16] experimented with transfer learning along with a fuzzy ensemble approach in deep learning for detection of breast cancer. They used the fuzzy ensemble approach that includes three models of CNN: VGG-11, ResNet50, and Inception v3 by using the modified Gompertz function to rank models. It achieved an accuracy of 98.98%. In the future, to ensure robust performance, they want to improve segmentation.

Dosovitskiy [18] and Google researchers, including Ashish Vaswani and Noam Shazeer, introduce Vision Transformer (ViT) by exploring the capabilities of transformers in the image classifications without CNNs. ViT achieves state-of-the-art accuracy across many models by pre-trained on large datasets and fine-tuning the small datasets. They want to work on ViT with CNN in their future research.

Zarif et al. [25] worked on pretrained models to create a hybrid model to detection and classification of breast cancer. Their purpose is to reduce manula errors in image analysis. They presented a hybrid model that is the combination of CNN and EfficientNetV2B3 with an accuracy of 96.3% to solve the problem of manula errors.

To address the problem of misdiagnosis rates in breast cancer screening, Tan et al. [26] introduce the RCM-YOLO network integrated with residual asymmetric dilated convolution, adaptive selection mechanisms, and cross-layer attention to optimize detection performance and achieve a map of 90.34%. They want to increase the accuracy in the future and improve the model to detect full marginal area of the mass so it can give better results.

Nadkarni and Noronha [27] experimented with convolutional neural networks using ensemble learning for detection of breast cancer. Their model includes the Hungarian optimization algorithm to enhance the classification accuracy, achieving accuracy of 95.7%.

In their paper, O'Shea [28] presented on An Introduction to CNN's architecture. The ability of CNN in analysing images is discussed with the help of convolutional and pooling layers for feature extraction in classification problems. They mention that CNNs have some benefits in relieving the model complexity while retaining the performances and draw attention to issues such as computational intensity. Therefore, they suggest future work aimed at improving CNN resistance in image applications.

Collectively from these studies, we know that future research will focus on integrating CNN with ViTs to refine the feature extraction process. So we focused on hybrid models, which are combination CNNs with ViTs, to leverage both local and global feature extraction for the improvement in classification performance and interpretability in breast cancer diagnostics.

## III. METHODOLOGY

Our proposed solution includes a hybrid architecture approach that supports architectures such as CNN and ViT. This approach balances the model, including the strengths of CNNs for extracting local features and ViTs for global dependencies capturing. In addition, we have also used a few other architectures including InceptionV3, DenseNet, Xception, SEResnet, and CNN to perform comparative analysis. This comprehensively evaluates models' performance across various feature extraction techniques and architectures.

### A. DATASET

The dataset for breast cancer classification was taken from Mammogram CLAHE dataset from kaggle, consists of 10,000 labeled images divided equally into the two classes: benign and malignant where I have used 5,000 samples in each class. "Fig. 1" also presents an idea of benign and malignant mammogram images thereby giving a clear clue on how the two categories look like. In tissue analysis each picture records the appearances of a tissue sample and is helpful for differentiation between benign and malignant cases. The dataset explained is balanced and is easier to apply for binary classification tasks, therefore supporting research in medical imaging applications for early detection of cancer. The dataset consists of high-resolution mammography images with $512 \times 512$ pixel resolution for uniformity. Both Craniocaudal (CC) and Mediolateral Oblique (MLO) views are available, with MLO being especially useful in identifying crucial areas such as the upper outer quadrant and pectoral muscle. These varied view types improve diagnostic efficiency and enable correct classification.

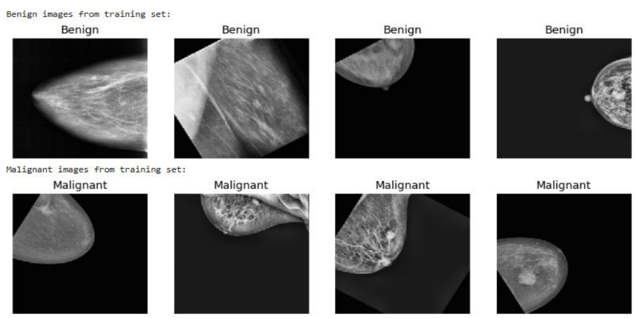


**FIGURE 1.** Benign and Malignant Mammogram images.

### B. INTRODUCTION TO CNN AND ViT MODELS

In Deep Learning, CNN is the most broadly used algorithm. CNN can be combined with other models to form a hybrid model to increase performance.

CNNs essentially solve complex image-based pattern recognition tasks, and with their accurate yet straightforward structure, it gives a more accessible route for introducing a novice to ANNs. These CNNs are a subset of deep learning models specifically designed for applications with images. They allow important features such as edges, textures, shapes, and even complex patterns to be learned in images through the employment of different layers of convolutional filters applied to the input.In other words, CNNs allow for the ability to capture spatial local relationships in images, making it especially useful for applications such as image classification, object detection, and analysis. In such a fashion, CNNs learn higher-level features progressively while reducing the spatial dimensions of data without losing important information, shows the training of Custom CNN; we got 0.85 as accuracy and a loss of 0.33. We can see that the single CNN model got very low accuracy and overfitted.

Vision Transformer(ViT) came out as an alternative to Convolution Neural Networks (CNN). Vision Transformer (ViT) which applies transformer architecture to classify images, originally it was designed for Natural Language Processing. ViT removes the dependency of extracting spatial features from image using convolutional layer by dividing the image into small, fixed size patches ($16\times16$), flattening them, processing them using a transformer.Each patch in the image is treated as a token, then positional embeddings are added inorder to get the spatial relationships.

Then the model applies the self-attention mechanisms across all patches, where we can capture the long-range dependecies and global context, which is more efficient that CNN. Compared to CNN, ViT requires fewer computational resources parallely maintaining higher accuracy, making the model more efficient for image recognition tasks.

### C. PROPOSED WORK:CNN+ViT MODEL

The proposed hybrid model integrates Custom Convolutional Neural Networks(CNN) and Vision Transformer(ViT-B16), this combination of CNN and ViT leverages the strengths of both the architectures which results in efficient performance over the image recognition field. Integrating both models, CNN can extract the features in early stages efficiently encoding the fine grained morphological patterns, while ViT could extract the loner range dependencies. This combination increases the robustness, the computational efficiency without letting the model underfit or overfit.

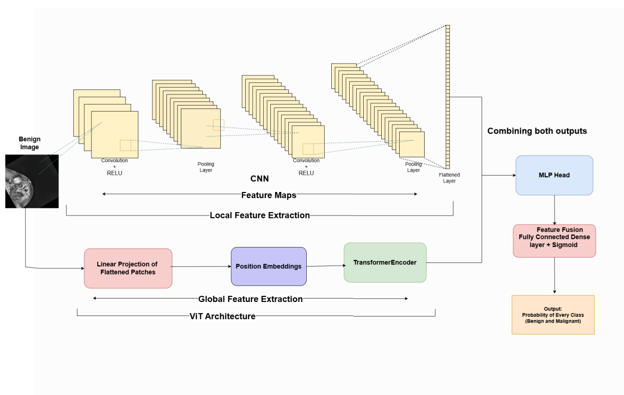


**FIGURE 2.** CNN+ViT architecture diagram.

"Fig. 2" demonstrate the complete architecture diagram of the CNN+ViT model.

#### 1) LOCAL FEATURE EXTRACTION VIA CNN

The convolutional layers perform correlation on the image and generate features maps that emphasize boundaries and small morphological features. These patterns are crucial for identifying of textures that may define areas that are benign or malignant.

The custom CNN is designed in a way which comprises of three convolutional layers to extract low level and middle level features. The layers are of

- **Conv Layer 1**: 32 filters, kernel size ($3 \times 3$), ReLU activation, followed by $2 \times 2$ MaxPooling.
- **Conv Layer 2**: 64 filters, kernel size ($3 \times 3$), ReLU activation, followed by $2 \times 2$ MaxPooling.
- **Conv Layer 3**: 128 filters, kernel size ($3 \times 3$), ReLU activation, followed by $2 \times 2$ MaxPooling.

And the final output is flattened to make it a 1D vector of local features for the subsequent fusion.

It moves across the image locally, and produces multiple feature maps and it is also for each filter that highlights various aspects for the same (edge detection, corner detection). Later, the learned features from deeper convolutional layers which are of higher order and complex features such as grouping of distorted cells. These layers are useful to have when looking for micro-calcification or distortions in the breast tissue. The ReLU activation function adds a certain non-linearity to the model, which makes the model in a position to comprehend the non-linear aspects of the data in addition to the linear aspects.

The feature maps are down sampled by Max Pooling layers to decrease the dimensionality of their spatial size but still retain any important features. This increases computational efficiency and also adds translation invariance, in which input image can be translated in the plane without affecting the likelihood that the pattern of interest appears in the image. In pooling layers, a pyramidal representation of features is developed by gradually reducing the spatial information with a focus on a few regions of the image. The convolutional and pooling layers create feature maps, at various levels of abstraction, where the initial levels detecting low level attributes like edges, fine structures or textures while the deeper levels detecting higher level attributes like shapes, sectors of interest etc.

This feature maps go through several convolution and pooling whereby they are converted to a single vector after being flattened. This operation unfolds the 2D spatial information into a usable format to connect with global feature maps from the Vision Transformer. Flattening led the local features extracted from all the regions of the image to be linearly disposed to be concatenated to global features. This step is an essential connection between CNN-based local representation feature and the proposed hybrid architecture. CNN is most optimal for tasks with an emphasis on fine detail of localized and small patterns pointing towards the irregular ones.

#### 2) GLOBAL FEATURE EXTRACTION VIA VISION TRANSFORMER (ViT)

The Vision Transformer improves the CNN architecture by learning the global relational feature of the input image. It processes the image at once, it takes into account relationship between distant regions. In the input image, the area of interest is split into patches of fixed size, where each size is a token of the transformer. The input image is resized to $224 \times 224$ pixels and patched to fixed size of non-overlapping patches of $16 \times 16$ pixels which amounts to 196 patches per image. These patches ensure that no part of the image remains unidentified, the entire image is viewed in terms of sequenced, manageable parts. This is particularly well suited in medical imaging because the abnormalities can be in small unknown areas of a large body. Each patch is flattens into 1D vector of length 768, positional embeddings are concatenated to preserve spatial relationships and linear layer is applied to patch to obtain the dense vectors. This increases pixel data with high dimensional feature vectors and latent feature space representation. The linear projection is useful as an embedding layer, such that each patch remains in the shape of a high-dimensional vector that can interact with other patches using the self-attention mechanism.

This encoded patch sequence is then fed into a stack of 12 transformer layers, each of which consists of 12 self-attention heads and a feed-forward multilayer perceptron (MLP) with an intermediate dimension of 3072 to capture relationships among patches and extract long-range dependencies and global emergent features. As the classification head is eliminated the ViT now acts as the global feature extractor. The decoder strategy allows the model to attend to specific regions of the image with regard to the overall relationship with other regions. For example, it is able to associate an irregularity or patch in a region of the image to another one located distant from the first one in the tissue map and thereby provides an overall idea of the structures of the tissue.

#### 3) CLASSIFICATION THROUGH FEATURE FUSION

Finally in the last stage the extracts from both CNN and ViT are incorporated very well to give a strong prediction.

The feature maps of the CNN we refer to as local features and that of the ViT as global features which are then joined end-to-end to form a new composite feature vector. Thus, the combined representation guarantees that the model will take into account both detailed and small-scale features and general and large-scale trends. The hybrid representation contain much information and allow for making the more accurate prediction of the model. For instance, local features may detect suspicious texture, but global features may reveal this texture as being significant in the total network of the tissue pattern.

To reduce the level of over-fitting a dropout layer is applied with dropout probability of 0.5 is used. This layer makes the corresponding neurons inactive during training but at times randomly does this allowing the model to learn better from unseen data.Then there is a fully connected dense layer with sigmoid activation that provides final probabilities for each class, malignant or benign.

This layer coordinates all feature extracted into a single decision making system. Another advantage is that the sigmoid activation function guarantees that the output is a probability distribution while the network's expectation directly indicates the confidence margin in its predictions.

### D. OTHER MODELS

In our experiment, we used several established architectures of deep learning such as DenseNet121, InceptionV3, XceptionNet, and SE-ResNet to compare and determine the effectiveness on the breast cancer classification dataset. Highly sophisticated architectures are known for advanced feature extraction techniques and state-of-the-art results in the classification of images, and this architecture provides insights about some strengths and weaknesses behind different design principles. Each of the models described was applied to our dataset of benign and malignant images, applying unique mechanisms to identify underlying patterns and ensure high classification performance. We have first tested with the custom CNN model, where it gave the training accuracy of 0.878 and validation accuracy of 0.853 as shown in "Fig.3", and the training loss is comparatively lesser that hybrid model, but the validation loss is more than the hybrid model.

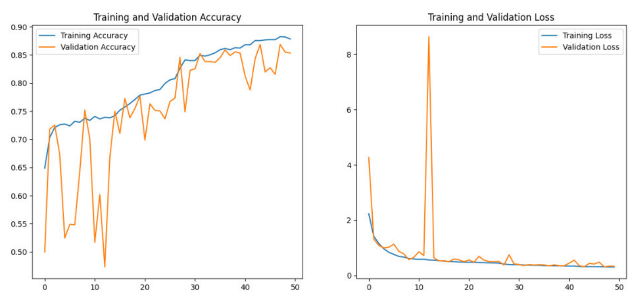


**FIGURE 3. Accuracy and loss curves of custom CNN.**

DenseNet121 [29] is an architecture that introduces dense connectivity which is that every layer is connected to every preceding layer. This encourages efficiency of feature reuse and minimizes redundancy. The network can easily extract the rich features and also ensuring computational efficiency. For our dataset, DenseNet121 succeeded in capturing the subtle differences between benign and malignant samples quite impressively, approaching good validation metrics. The low training loss and high validation accuracy of in this model, however, contributed to caution regarding the possibilities of overfitting we can see this clearly in "Fig. 4" - it might have so focused on training data and not generally enough on the unseen data.

InceptionV3 [30] brings the capability for realizing diverse patterns in the input images. Using inception modules

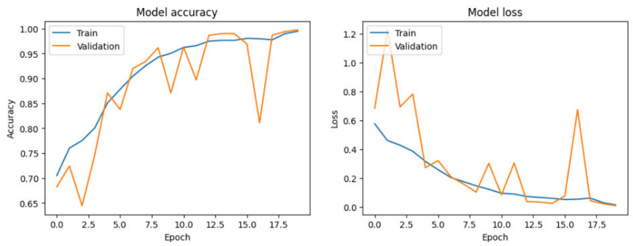


**FIGURE 4.** Accuracy and loss curves of DenseNet121.

comprising various filters of different sizes to extract features at multiple scales, While the architecture is highly efficient with factorized convolutions and auxiliary classifiers, the loss of validation is little. In our dataset, when applied, it very well highlighted the extraction of critical features. Its performance on the unseen datasets may not be strong which suggests that fine-tuning or data augmentation must be further used in order to improve generalization. Same as DenseNet, it was also overfitted with 0.99 accuracy, see the "Fig. 5".

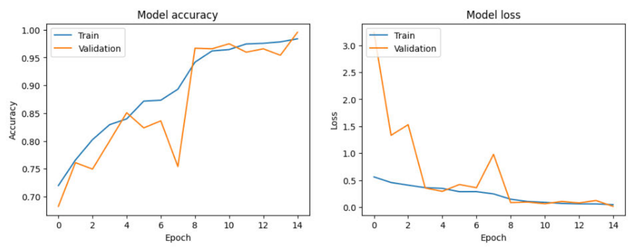


**FIGURE 5.** Accuracy and loss curves of InceptionV3.

SE-ResNet [31] adds a squeeze-and-excitation blocks to the baseline of the ResNet architecture which recalibrate the feature maps to emphasize more over the relevant patterns. This mechanism clearly improves upon the representational power of the network, with clear suitability to datasets where feature distinctions are subtle. On our dataset, SE-ResNet presented robust results, fairly balancing the training and validation performance. Unlike some of the other architectures, SE-ResNet showed a better capability of generalization, hence showing the ability to perform well across a wider range of clinical scenarios of medical imaging. Training and validation curves of SE-ResNet was shown in "Fig. 6".

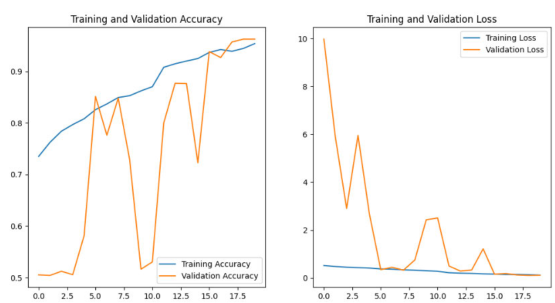


**FIGURE 6.** Accuracy and loss curves of SE-ResNet.

XceptionNet [32] is an extension which is based on depth-wise separable convolutions that are efficiently separate spatial and also channel-wise feature extraction for a very lightweight architecture. They can capture fine-grained details of the image. When used for the dataset, XceptionNet gave very close to perfect accuracy we can see this in the "Fig. 7", effectively classify benign tumor images from malignant tumor images. On other metrics, the model's performance seems to hint at overfitting, meaning it could face a serious problem of generalization under conditions brought on by real-world variability.

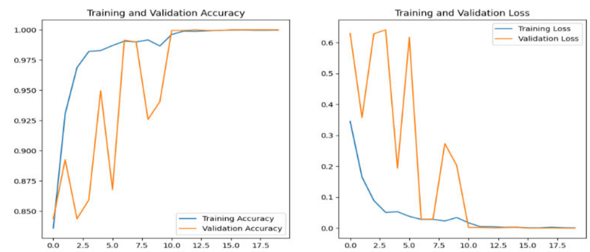


**FIGURE 7.** Accuracy and loss curves of XceptionNet.

## IV. RESULTS

In our experimental results of the CNN-ViT model which highlights the effectiveness in breast cancer classification, balancing accuracy and generalization. As shown in the "Fig. 8", the validation accuracy is 0.901 and the training accuracy is 0.853 for 20 epochs. With such accuracy, the model performs well without any overfitting, as indicated by a relatively low gap between training and validation metrics. This accuracy is obtained by combining CNN's capacity to capture local features which are the edges, and textures, and ViT's ability to capture the global dependencies across the image. The model's losses are 0.323 and 0.260 in training of the model and validation respectively. These results indicate that the CNN+ViT approach successfully exploits the complementary advantages and uses of CNNs and ViTs to create a model that can generalize well to new data. This one mainly refers the importance in medical screening, where robustness is crucial.The confusion matrix of our study, CNN+ViT is presented in "Fig.9", proving that the approach efficiently

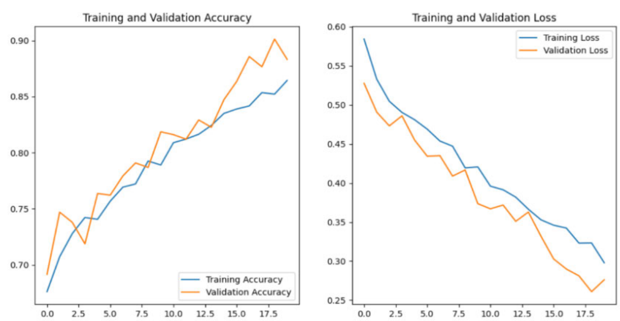


**FIGURE 8.** Training and validation graphs of CNN+ViT model.

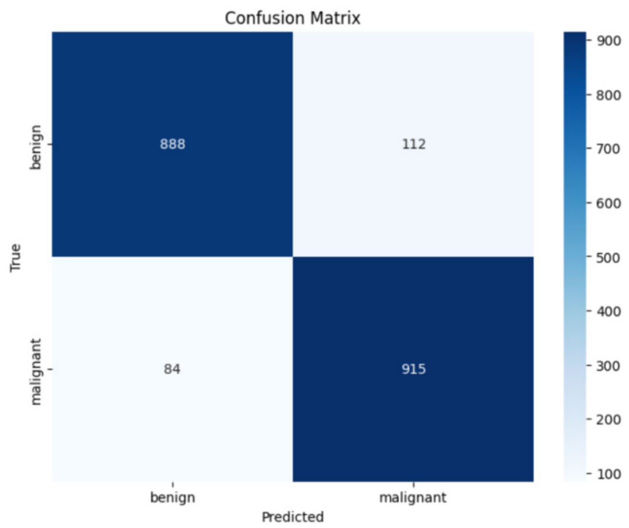


**FIGURE 9.** Confusion matrix of CNN+ViT model.

selects mammogram images with benign and malignant characteristics with a greater degree of accuracy. The quantity of false positive and false negatives has been minimal and this shows that the model is reliable.

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| benign | 0.91 | 0.89 | 0.90 | 1000 |
| malignant | 0.89 | 0.92 | 0.90 | 999 |
| accuracy | | | 0.90 | 1999 |
| macro avg | 0.90 | 0.90 | 0.90 | 1999 |
| weighted avg | 0.90 | 0.90 | 0.90 | 1999 |

**FIGURE 10.** Classification report showing precision, recall, and F1-score for benign and malignant classes in CNN+ViT model.

A classification report "Fig.10" provides detailed metrics including precision, recall, and F1-score for both benign and malignant cases. For benign cases, a precision of 0.91 and recall of 0.89 were observed. For malignant cases, precision was 0.89 and recall reached 0.92. The classification is effective across both categories, with F1-scores of 0.90 and an overall accuracy of 90%. Macro and weighted averages of all metrics also yielded a consistent 0.90, indicating balanced performance across classes.

The model demonstrates strong effectiveness through accurate classification and maintains equally optimized performances for both benign and malignant cases. Data analysis requires special attention in medical contexts because both incorrect positive and negative results create severe medical issues.

## V. DISCUSSION

The results in the below table "Fig. 11", are from the other pre-trained models, base models and hybrid model which include, classic CNN, Hybrid (CNN Combined with ViT), and pre-training models DenseNet121, Inception V3, XceptionNET, and SE-ResNet. Therefore, the best result in terms of accuracy corresponds to the minimum loss: 0.238 with all models used jointly, CNN + ViT. It has shown training accuracy 85.3% and validation accuracy 90.1% and validation loss of 0.260 we can see this in the "Fig. 8". The difference in aggressiveness between the training and validation sets gives us confidence that the hybrid model does not over-fit on the training data and can therefore be trusted more in the real world. On the other hand, although pre-trained models achieve high metrics, the model exhibits overfitting. For instance, DenseNet121 achieves a near-perfect validation accuracy of 99.7% with extremely low losses (training loss: 0.015, validation loss: 0.009). Likewise, validation accuracy provided by Inception V3 and XceptionNET models are 99.5% and 100% while the validation loss for XceptionNET model is 0. While these results could be impressive on the practical level, such performance may indicate that these models simply remember training data, rather than recognizing general, potent patterns.

| MODEL | Training Accuracy | Validation Accuracy | Training Loss | Validation Loss |
|---|---|---|---|---|
| CNN | 0.878 | 0.853 | 0.306 | 0.332 |
| Hybrid(CNN+ ViT) | 0.853 | 0.901 | 0.323 | 0.260 |
| DenseNET121 | 0.994 | 0.997 | 0.015 | 0.009 |
| Inception V3 | 0.983 | 0.995 | 0.047 | 0.016 |
| XceptionNET | 1.00 | 1.00 | - | - |
| SE - ResNet | 0.954 | 0.962 | 0.114 | 0.106 |

**FIGURE 11.** Comparision table with all models.

**TABLE 1.** Comprehensive performance comparison of models using multiple metrics.

| Model | Precision (%) | Recall (%) | F1-Score (%) |
|---|---|---|---|
| CNN | 51.00 | 51.00 | 50.00 |
| CNN + ViT (Hybrid) | 90.00 | 90.00 | 90.00 |
| SE-ResNet | 96.00 | 96.00 | 96.00 |
| XceptionNet | **100.00** | **100.00** | **100.00** |

*Note:* DenseNet121 and InceptionV3 showed performance comparable to XceptionNet, with F1-scores close to 99%, and were excluded from the table for clarity.

"Table. 1" demonstrates clinical metrics assessment of models through Precision, Recall and F1-Score evaluation to provide strong metrics analysis beyond accuracy measures. The Hybrid model integrates CNN and ViT to obtain better results than the baseline CNN model with a 90% F1-score. XceptionNet demonstrates the best performance among all models because it achieves 100% precision, recall, and F1-score metrics. The perfect results of XceptionNet are similar to its validation accuracy. Still, its performance indicates potential overfitting issues ("Fig. 7"). The paper includes an extensive analysis of this matter for its future research direction.

Therefore the baseline CNN with validation accuracy of 85.3% and validation loss 0.332 is quite good but inferior to the hybrid model. In conclusion, based on the comparing the generalization performance and the accuracy of the breast cancer, the hybrid model CNN+ViT is proved to be the most reasonable.

## VI. CONCLUSION AND FUTURE SCOPE

In summary, the main aim is to develop a robust breast cancer classification system by utilizing the CNN+ViT architecture combining the integrating model CNN for the detailed extraction of features locally and the ViT to capture the global contexts; this methodology improves the classification accuracy but also preserves the generalization capabilities. The CNN+ViT model shows good performance with balanced training and validation accuracy, showing the capability to differentiate benign from malignant cases with minimal overfitting. Model architecture optimization could further improve the computational efficiency of the proposed model for future deployment in clinical environments.

Further, future research will include the comparative analysis of computational efficiency factors like inference time, memory consumption, and Flops in all the models tested in order to determine their applicability for real-time clinical use. Further, we also plan to study the abnormally high validation accuracy of the XceptionNet model, which can be a pointer towards overfitting or data leakage. To further comprehend model behavior, we intend to use interpretability methods like Grad-CAM and t-SNE to provide visualizations of feature importance and verify that the network is indeed learning significant diagnostic patterns instead of artifacts.

Finally, increasing the dataset from a sample size of 5,000 to lakhs of samples can include several histopathological images including diverse samples of tissues in addition. The adaptability and diagnosis of the developed model are improved further on this addition. Improving interactivity interface or cloud-based application helps real-time access by health professionals, improves early detection, and leads to an increase in the outcome for a patient.

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